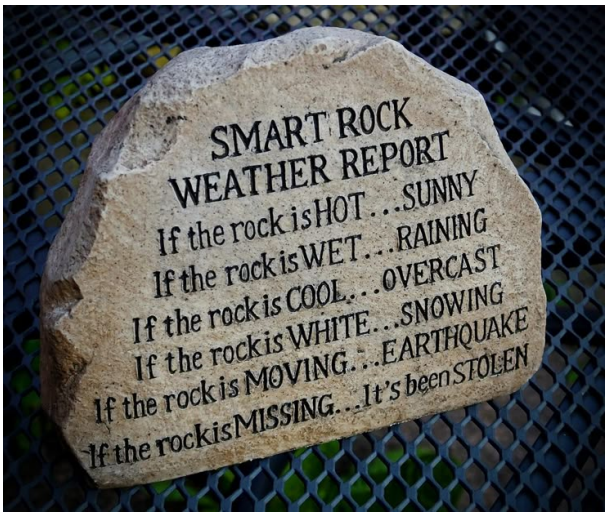
Science Olympiad
Meteorology B Key
Greenbrier Invitational



Team Name: _____**KEY**_____

Team Number: _____

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Directions:

A. You will have 50 minutes to complete this test

B. Ties will be broken based on correctness of the following questions: 6, 7, 4, 1

C. Some questions are supposed to be difficult, so don't worry if you aren't able to finish all the questions.

Page:	2	3	4	5	6	Total
Points:	9	10	11	8	8	46
Score:						

1. (2 points) Order the following layers of the atmosphere from lowest to highest altitude:
   - Mesosphere
   - Troposphere
   - Exosphere
   - Thermosphere
   - Stratosphere

Solution: Lowest: Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere: Highest.

All or nothing for points.

2. (2 points) Which layer of the atmosphere is the most dense, and why?

Solution: Troposphere, since it's compressed by the layers above it.

3. (3 points) For each layer of the atmosphere, what is the most common pollutant?

Solution: Troposphere: Particulate Matter (PM) (1pt); Stratosphere: Ozone Depleting Substances (ODS) (1pt); Mesosphere/Thermosphere/Exosphere: None (1pt).

One point

4. (2 points) How much does the sun's longitude change with respect to the equator each day?

Solution: Accept any answer in the range $0.39° - 0.41°$

5. (2 points) Using your answer from the previous question, what would the longitude of the subsolar spot be on March 30th? Show your work.

Solution: The spring equinox is March 20th. March 30th is 10 days later.
10 days $\times$ $0.4°$ = $4°$.
In the spring, the sun is moving north, making the final position $4°N$. All or nothing points.

For question 6, write the appropriate instrument to monitor the atmospheric condition. There may be more than one correct answer, but there will be only one best answer.

6. (6 points) Air temperature, atmospheric pressure, relative humidity, wind speed, wind direction, precipitation amount, cloud base height, solar radiation intensity, atmospheric ozone concentration, particulate matter, vertical temperature profile, and atmospheric water vapor content.

Solution: Air temperature (surface) — Thermometer
Atmospheric pressure — Barometer
Relative humidity — Hygrometer
Wind speed — Anemometer
Wind direction — Wind vane
Precipitation amount — Rain gauge
Cloud base height — Ceilometer
Solar radiation intensity — Pyranometer
Atmospheric ozone concentration — Dobson spectrophotometer
Particulate matter ($PM_{2.5}$ / PM_{10}) — Air quality monitor
Vertical temperature profile — Radiosonde
Atmospheric water vapor content — Microwave radiometer.
Award 0.5 points per correct

7. (2 points) Define and explain the meteorological significance of a wet-bulb temperature

Solution: Wet-bulb temperature is the lowest temperature air can reach through evaporative cooling alone, under constant pressure. (1pt)
It is important since it helps with the estimation of humidity, cloud formation, etc. (1pt)

8. (3 points) At the wet-bulb temperatures $0°C, 10°C$, and $20°C$, what cloud formations would be expected

Solution: $0°C$: Nimbus/Stratonimbus (1pt) $10°C$: Cumulus (1pt) $20°C$ Cumulonimbus (1pt)

9. (2 points) What is the most important meteorological condition in order for a green flash to happen, and why?

Solution: A temperature inversion (1pt), because refraction and dispersion of sunlight is enhanced.

10. (2 points) Compare and contrast mercury-based and alcohol-based thermometers based on their effectiveness at different temperatures

Solution: Mercury-based thermometers are better at measuring higher temperatures (1pt), whereas alcohol-based thermometers are better at lower temperatures (1pt).

11. (2 points) Which region of the world has the most virga, and why?

Solution: The Great Basin (Nevada, US) (1pt), because of the high-elevation basins, extremely dry lower troposphere, frequent mid-level clouds, and precipitation aloft

12. (2 points) If the Coriolis effect magically disappeared immediately, what would be expected over the course of the next five minutes?

Solution: If the Coriolis effect disappeared, surface winds would blow directly from high to low pressure without curving, trade winds and westerlies would vanish, cyclones and hurricanes could not rotate, ocean gyres would collapse, and precipitation and climate patterns worldwide would be drastically altered.

Note to graders: this is supposed to be a more fun question before the more difficult scenario-based analysis ones, so please grade easier on this one.

For questions 13-16, use the following information: On a summer afternoon, a weak surface low and a nearby stationary front allow warm, moist air to pool over a flat inland region. Daytime solar heating increases instability, producing scattered cumulus that grow into isolated thunderstorms, but weak wind shear prevents storm organization. The storms produce brief heavy rain and occasional lightning, with no hail or damaging winds.

13. (1 point) Identify the lifting mechanism responsible for thunderstorm initiation in this scenario.

Solution: Surface heating (convective lifting), aided by weak convergence along the stationary front.

14. (2 points) Explain why the thunderstorms remain nonsevere despite the presence of warm, moist air.

Solution: Although instability is present, weak vertical wind shear prevents storm organization and long-lived updrafts, limiting storm intensity.

15. (2 points) Describe how the vertical wind shear profile influences storm type and lifetime in this situation.

Solution: Minimal vertical wind shear favors single-cell (air-mass) thunderstorms, where the downdraft quickly cuts off the updraft, resulting in short-lived storms.

16. (3 points) If mid-level dry air were introduced into this environment, analyze how precipitation type and downdraft strength might change without necessarily producing severe storms.

Solution: Dry mid-level air would enhance evaporational cooling, strengthening downdrafts and increasing gustiness, while precipitation may partially evaporate as virga or produce brief heavy rain, still without sufficient shear to become severe.

Use the following information to answer questions 17-20: As the single-cell thunderstorms develop, they produce brief, localized heavy rain and occasional lightning strikes, but most precipitation evaporates before reaching the ground in some areas. Gusty downdrafts create temporary wind shifts at the surface, and minor puddling occurs in low-lying areas, but no flooding develops. The storms quickly dissipate as the updraft collapses, leaving behind scattered clouds and a temporary drop in temperature.

17. (1 point) Identify what meteorological phenomenon is observed when precipitation evaporates before reaching the ground.

Solution: Virga

18. (2 points) Explain why surface gusts occur during these short-lived thunderstorms.

Solution: Gusts occur because downdrafts of cooler air from collapsing updrafts accelerate toward the surface, creating brief, localized wind events.

19. (2 points) Describe how uneven rainfall distribution in this scenario could impact local areas differently.

Solution: Some areas may receive brief heavy rain, causing temporary puddles, while others experience little or no surface rainfall due to virga, leading to unequal soil moisture and minor localized flooding.

20. (3 points) Analyze how the combination of virga and downdrafts affects near-surface temperature and humidity immediately after the storm.

Solution: Evaporation of falling precipitation (virga) cools and moistens the air aloft, while downdrafts push cooler, denser air to the surface, producing a temporary drop in temperature and a brief increase in local wind gusts, often followed by slightly increased surface humidity as the storm dissipates.